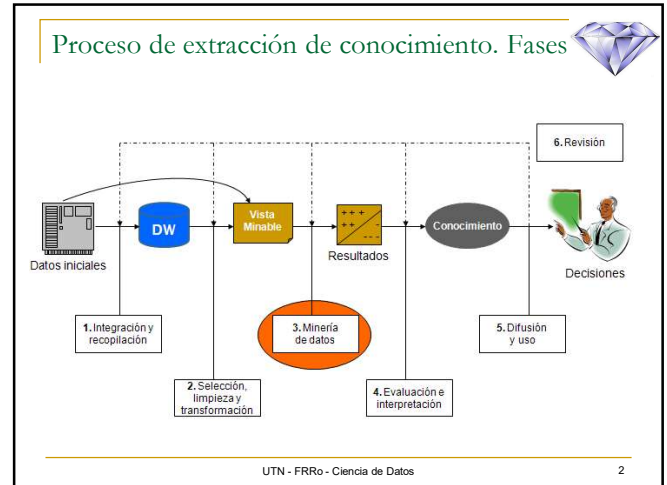
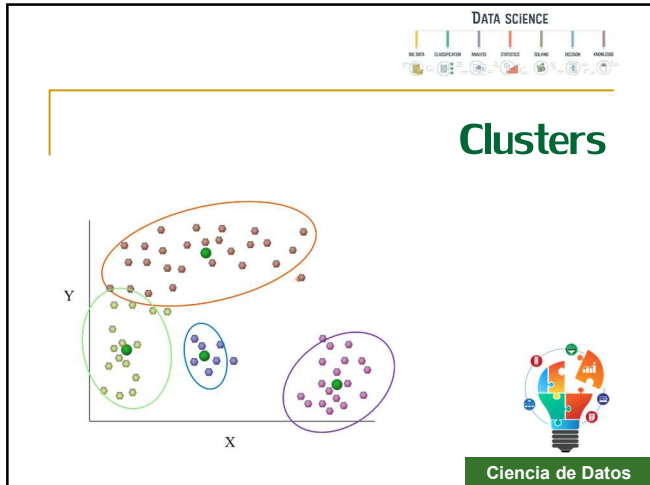




KDD: Tareas y técnicas



Técnicas	Tareas		Predictivos (supervisados)				Descriptivos (no supervisados)		
	Clasificación	Predicción			agrupamiento	Asociación		Otros	
Redes Neuronales	X	X			X				
Arboles de Decisión	X	X			X	X			
Kohonen					X				
Regresión paramétrica		X							
Regresión Logística	X					X			
k means / k medioides / Jerárquicos					X				
A priori						X			
Análisis factorial								X	
Análisis Discriminante	X								
Algoritmos genéticos	X	X			X	X		X	
Naive Bayes	X								
Vecinos mas próximos	X	X			x				

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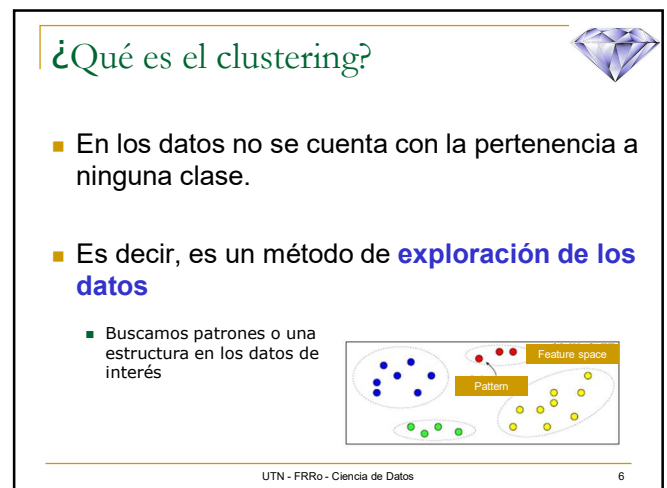
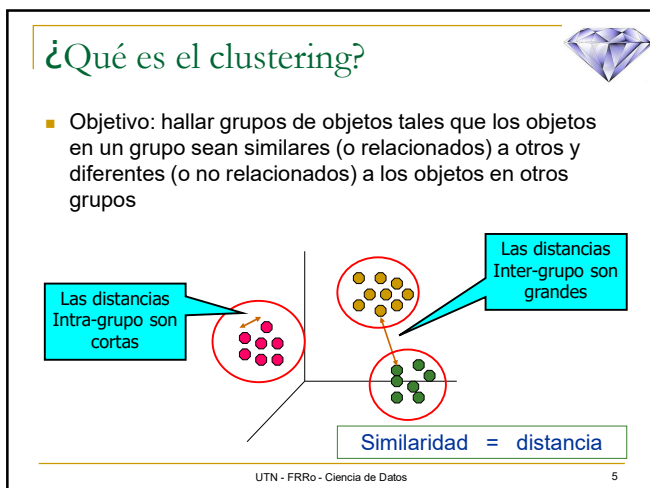
3

El conjunto de datos

- *El conjunto de datos*
 - Los datos son *observaciones de algún proceso*.
 - Cada observación consiste de *p variables medidas*
 - Agrupadas en *vectores filas*

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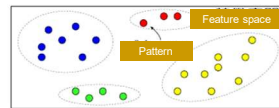


¿Qué es el clustering?

■ Una forma de aprendizaje no supervisado

- No se tienen ejemplos de cómo deben ser agrupados

Clustering = agrupamiento



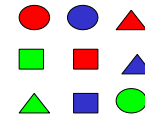
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Representación de los patrones

■ ¿Qué variables características usar?

- Normalmente, ninguna guía teórica que sugiera los patrones apropiados y las características a usar en una situación específica
- El usuario generalmente debe proporcionar esta guía
- Un análisis cuidadoso de las características disponibles puede conducir a mejores resultados

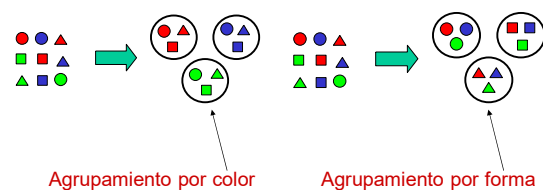


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Representación de los patrones

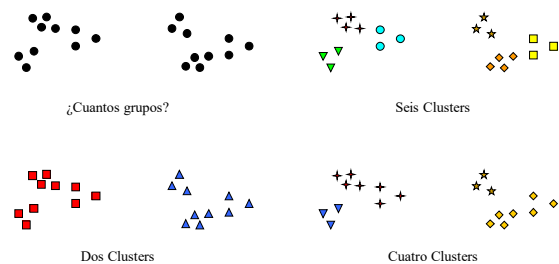
■ Dos agrupamientos para unos mismos datos



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La noción de grupo puede ser ambigua



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Tipos de Clustering

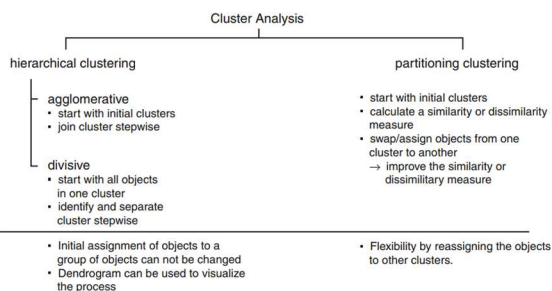


Fig. 7.4 Overview of Clustering algorithms

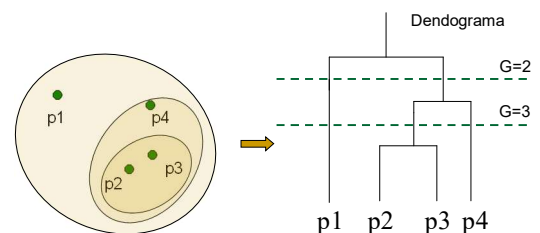
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Tipos de Clustering

Clustering jerárquico

- Un conjunto de clusters anidados organizados como un árbol jerárquico con un cluster único arriba, agrupando todos los datos, y clusters con un solo elemento abajo

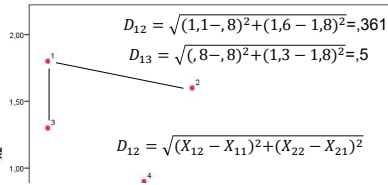


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Cluster Jerárquico

	X1	X2
1	,80	1,80
2	1,10	1,60
3	,90	1,30
4	1,00	,90
5	1,40	,60
6	1,50	,10
7	1,10	,10



Caso	distancia euclídea						
	1	2	3	4	5	6	7
1	,000	,361	,500	,922	1,342	1,838	1,726
2	,361	,000	,424	,707	1,044	1,552	1,500
3	,500	,424	,000	,447	,922	1,389	1,237
4	,922	,707	,447	,000	,500	,943	,806
5	1,342	1,044	,922	,500	,000	,510	,583
6	1,838	1,552	1,389	,943	,510	,000	,400
7	1,726	1,500	1,237	,806	,583	,400	,000

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Cluster Jerárquico

	X1	X2
1 y 2	,80	1,80
3	1,10	1,60
4	,90	1,30
5	1,00	,90
6	1,40	,60
7	1,50	,10

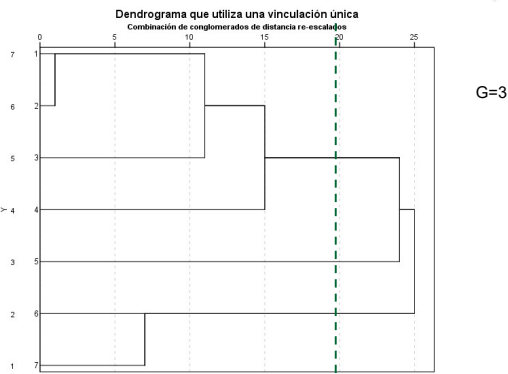
Matriz de distancias

Caso	distancia euclídea					
	1	3	4	5	6	7
1	,000	,500	,922	1,342	1,838	1,726
3	,500	,000	,447	,922	1,389	1,237
4	,922	,447	,000	,500	,943	,806
5	1,342	,922	,500	,000	,510	,583
6	1,838	1,389	,943	,510	,000	,400
7	1,726	1,237	,806	,583	,400	,000

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Cluster Jerárquico



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Criterios de aglomeración

Table 7.24 Hierarchical clustering algorithms

Name of algorithm	Characteristics
Single-linkage	Smallest distance between objects is used. All proximity measures can be used. Outliers remain separated—identification is possible. Groups that are close to each other are not separated.
Complete-linkage/maximum distance	Largest distance between objects is used. All proximity measures can be used. Outliers are merged to clusters—identification not possible. Tends to build many small clusters.
Average linkage/average distance/within groups linkage	The average of the object distances between two clusters is used. All possible object distances will be taken into account. All proximity measures can be used.
Centroid/average group linkage/ between groups linkage	Similar to centroid, but the distances between cluster centers are taken into account. The compactness of the clusters is also relevant. Squared Euclidean distance is used.
Median	Similar to centroid, but the cluster centers are determined by the average of the centers, taking the number of objects per cluster into account (number of objects is known as the weight). Only distance measures can be used.
Ward's method	Cluster centers are determined. The distances between all objects in each cluster and the center are determined and cumulated. If clusters are merged, then the distance from the objects to the new cluster center increases. The algorithm identifies the two clusters where the distance increment is the lowest. See Buhl (2012, p. 650). Cluster sizes are approximately the same. Only distance measures can be used.

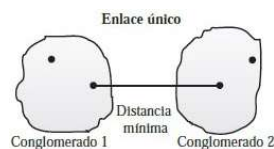
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Criterios de aglomeración

Encadenamiento Simple

La distancia entre grupos se define como la del par de individuos que está más cercano



Encadenamiento Completo

La distancia entre grupos la definimos ahora como la mayor distancia entre pares de individuos, uno de cada grupo.



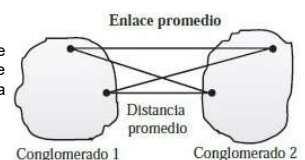
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Criterios de aglomeración

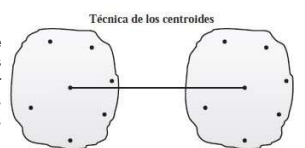
Encadenamiento Promedio

La distancia entre dos clusters se define como la distancia media entre todos los pares de individuos de cada grupo.



Método del Centroide

Los grupos una vez formados se representan por sus valores medios para cada variable, es decir su vector de medias y las distancias entre los grupos se definen en términos de las distancia entre vectores de medias.



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Criterios de aglomeración

Método de Ward

- Se determinan los centros de los clusters y las distancias entre los individuos y el centro de cada cluster.
- Si se fusionan los grupos, la distancia desde los objetos hasta el nuevo centro del grupo aumenta. El algoritmo identifica los dos grupos donde el incremento de distancia es menor.



Criterios de aglomeración

La diferencia entre los distintos métodos está en la **diferente forma** de definir la **distancia** entre dos grupos de individuos.

Medidas de Proximidad

Variables binarias → Coeficiente de Tanimoto

Variable	Special offer last month	Customer satisfied	Supported by manufacturer	Offered guarantee >1 year	Recyclable packaging
Product A	1	0	1	1	1
Product B	1	1	0	1	0
Product C	0	1	0	0	1

Tabla de Contingencia

		Object 2	
		1	0
Object 1	1	a	b
	0	c	d

$$s_{ij} = \frac{a}{a + b + c}$$

$$s_{AB} = \frac{2}{2 + 2 + 1} = 0.4$$

Medidas de Proximidad

Distancia Euclídea

$$D_{ij} = \sqrt{\left(\sum_{k=1}^p (X_{ik} - X_{jk})^2\right)}$$

✓ Es la medida usada más frecuentemente...

✓ No soporta cambios de escala

Medidas de Proximidad

Distancia Euclídea estandarizada

$$D_{ij}^2 = \left(\frac{\sum_{k=1}^p (X_{ik} - X_{jk})^2}{s_k} \right)$$

s_k es la desviación típica de la variable k-ésima

✓ Invariante al cambio de escala

Medidas de Proximidad

Distancia de Mahalanobis

$$D_{ij} = \sqrt{(X_i - X_j)' * S^{-1} * (X_i - X_j)}$$

S es la matriz de covarianzas dentro del grupo

✓ Invariante al cambio de escala

Clustering

Definir medidas de proximidad

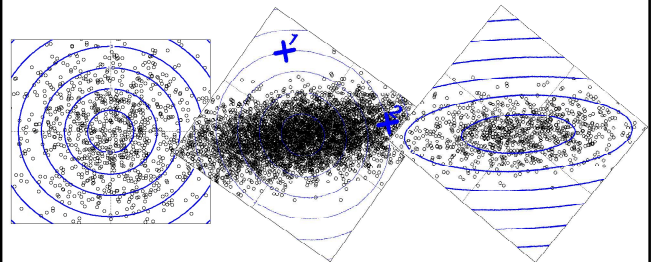
Tanimoto
Euclídea
Euclídea Estandarizada
Mahalanobis
Etc.

Definir Criterio de Aglomeración

Encadenamiento Simple
Encadenamiento Completo
Encadenamiento Promedio
Método del Centroide
Método de Ward
Etc.

Euclidean vs. Mahalanobis

Contours of equidistant points from the origin

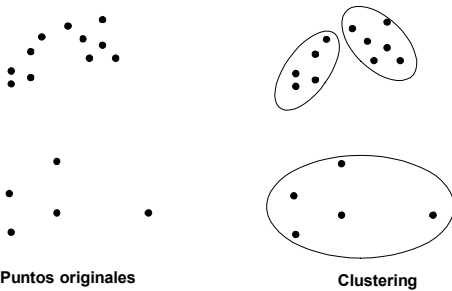


Puntos distribuidos uniforme/, Euclidean distance Puntos distribuidos Normal/, Euclidean distance Puntos distribuidos Normal/, Mahalanobis distance

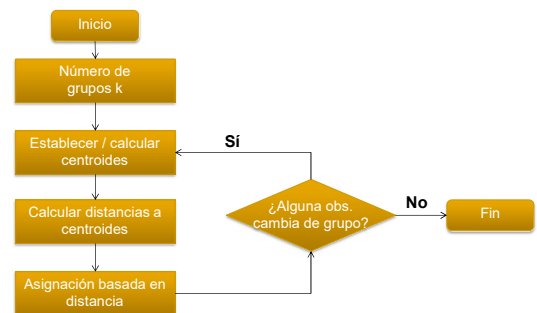
Tipos de Clustering

Clustering no jerárquico

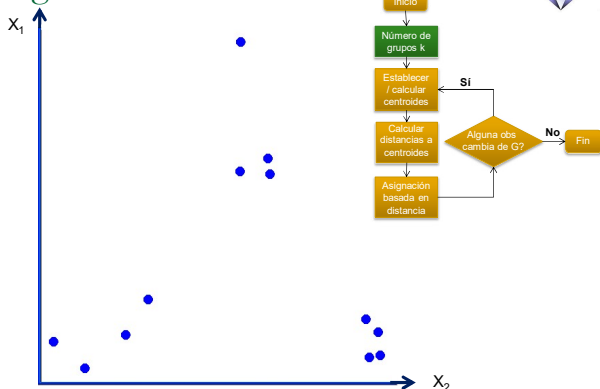
- División de los datos en subconjuntos (clusters) en una partición en un único nivel (no anidado)



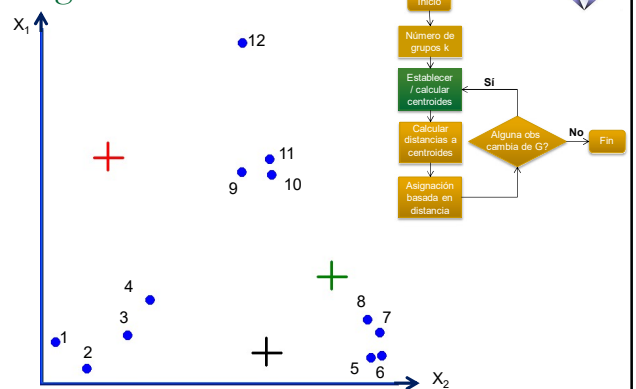
Algoritmo k-Medias

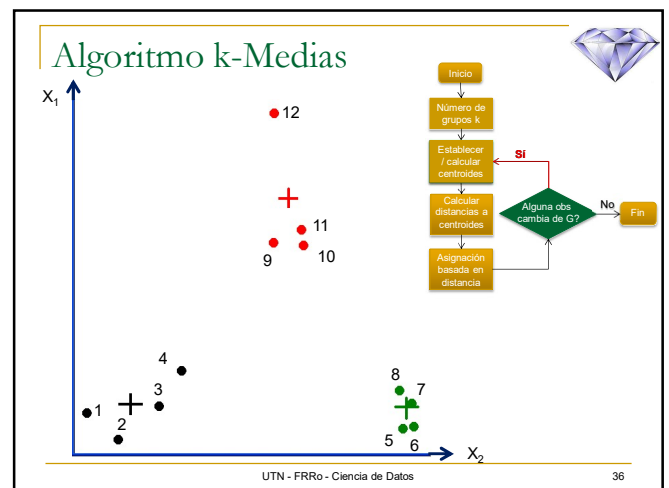
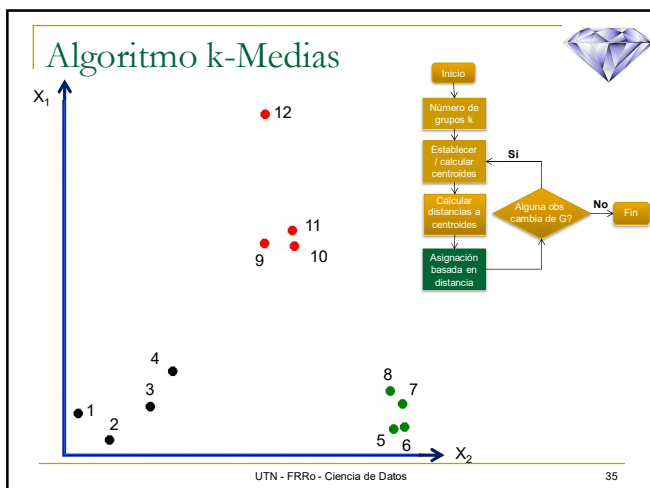
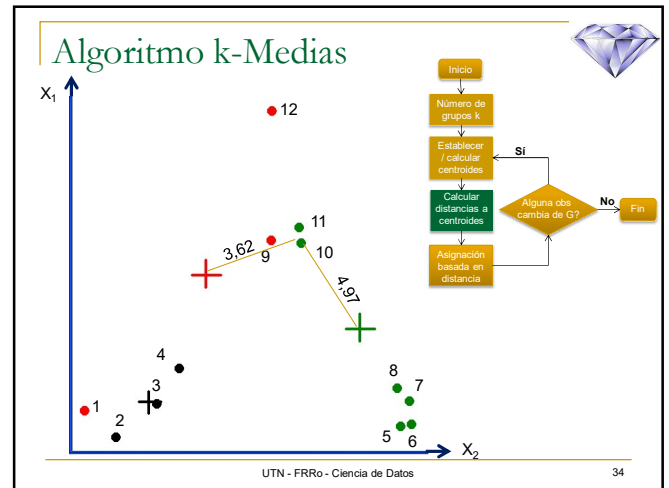
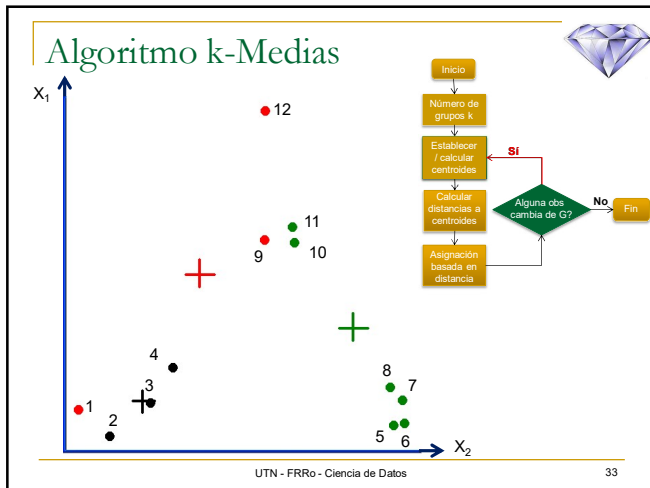
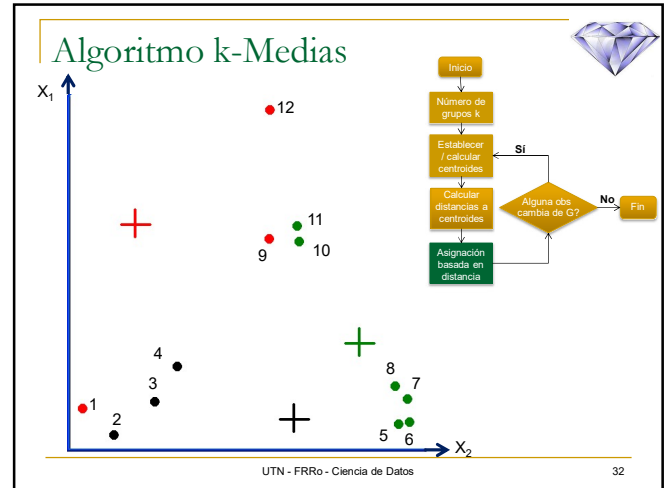
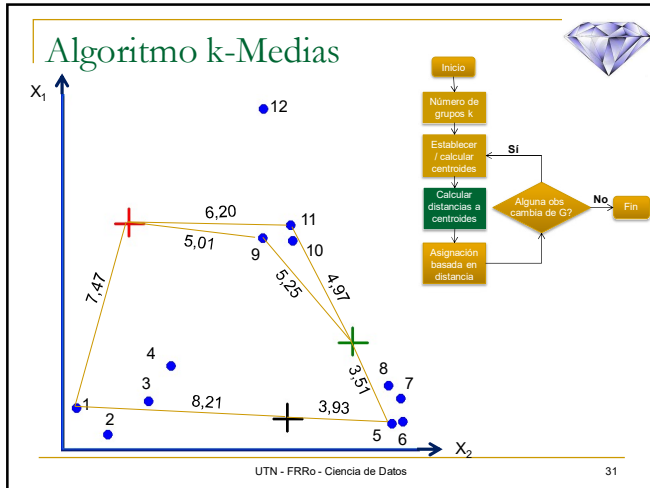


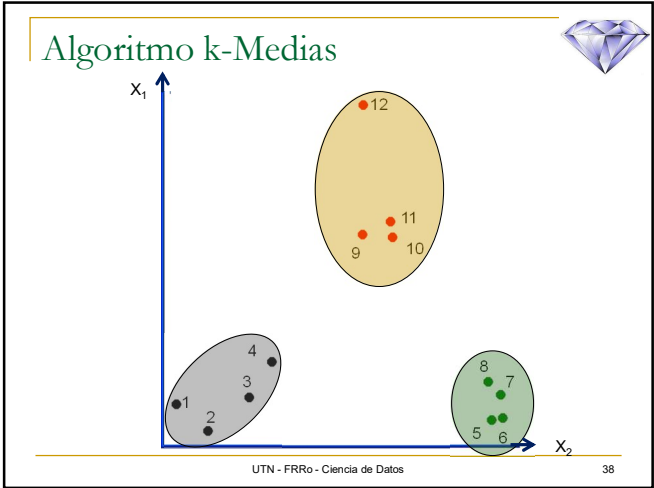
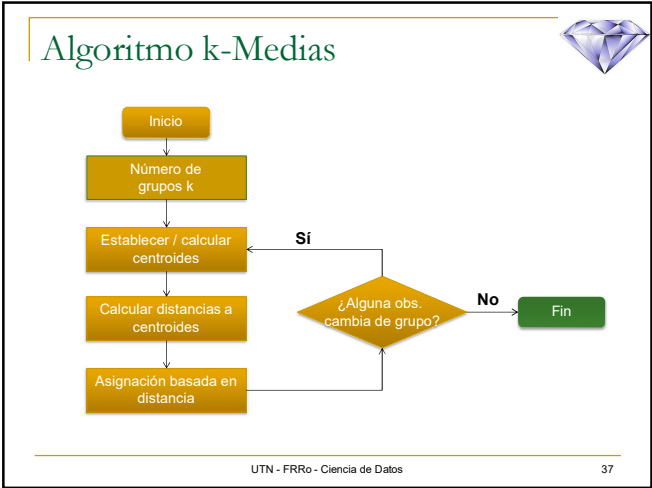
Algoritmo k-Medias



Algoritmo k-Medias







Cluster Jerárquico – Beer.sav

■ Variables Numéricas

		Grupos							
Media Población	Variables	2		3		4			
	Precio								
	Costo								
	Calorias								
	Sodio								
	Alcohol								
		G1	G2	G1	G2	G3	G1	G2	G3 G4

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Cluster Jerárquico – Beer.sav

■ Variables Categóricas

		Grupos							
Variables		2		3		4			
Origen									
Disponibilidad									
Light									
		G1	G2	G1	G2	G3	G1	G2	G3 G4

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¿Cuántos Clusters?

Table 7.25 Methods to determine the number of clusters

$\sqrt{\text{number of objects}}$	In reference to Marila et al. (1979), this rule of thumb approximates the number of clusters.
Elbow-criterion	As explained in Sect. 6.3.2 and Fig. 6.18, the dependency of a classification criterion vs. the number of clusters can be visualized in a 2D-chart. In the case of PCA/PFA, this is called a scree plot. In cluster analysis, at the vertical axis the sum of squares of distances/ errors can be used. In reverse, the percentage of variance/information explained can be assigned to the vertical axis. See Figs. 7.12 and 7.13. An example can be found in Exercise 5 of Sect. 7.4.3.
Silhouette charts	To measure the goodness of a classification, the silhouette value S can be calculated: 1. Calculation of average distance from the object to all objects in the nearest cluster. 2. Calculation of average distance from the object to all objects in the same cluster the object belongs to. 3. Calculation of the difference between the average distances (1)-(2). 4. This difference is then divided by the maximum of both those average distances. This standardizes the result. $s_i = \frac{\max(1 - d_{i,j}, d_{i,j} - d_{i,k})}{\max(d_{i,j}, d_{i,k})}$ More details can be found in Streij et al. (1997, pp. 5-7). S can take on values between -1 and $+1$. If $S = -1$, the object is not well classified. If $S = 0$, the object lies between two clusters, and if $S = +1$ the object is well classified. The average of the Silhouette values of all objects represents a measure of goodness for the overall classification. As outlined in IBM (2015b, p. 77) and IBM (2015b, p. 209), the IBM SPSS Modeler calculates a Silhouette Ranking Measure based on the Silhouette value. It is a measure of cohesion in the cluster and separation between the clusters. Additionally, it provides thresholds for poor (up to +0.25), fair (+0.5), and good (above +0.5) models.
Information criterion approach	Information criteria such as the Bayesian Information Criterion (BIC) or the Akaike Information Criterion (AIC) are used to determine the appropriate number of clusters. For more details, see Tavani (2013, pp. 61-63).

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¿Cuál método usar?

Table 7.1 Characteristics of clustering algorithms by category

Hierarchical clustering (TwoStep)	Pros: – The number of clusters must not be defined in advance. – Clusters are defined stepwise. The order of clusters defined can be visualized in a tree. See explanation of dendrogram (Fig. 7.10). – Clustering results are deterministic (non-random). For example, the result does not depend on the order of the objects/data records analyzed in the dataset. – The quality of the solution can be determined by using problem-specific distance functions (single-linkage, complete-linkage, etc., in the case of agglomerative algorithms). Cons: – Assignment of the objects to a specific cluster can never be changed/ revised. – It is assumed that continuous variables are normally distributed. Transformation is necessary in advance. See Sect. 3.2.5.
Partitioning clustering (K-means)	Pros: – More flexible than agglomerative algorithms, based on the reassignment of the objects to other clusters. – Non-normally distributed variables can be used. Cons: – The number of clusters must be defined. – Initial clustering is often based on heuristic methods, but initial clusters determine the quality of the final solution. – The order of cluster definition can't be visualized in a tree (called flat-clustering).

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Bibliografía



- Data Mining with SPSS Modeler. Capítulo 7.

